

Training Loss Convergence of the Residual U-Net Inverse PINN

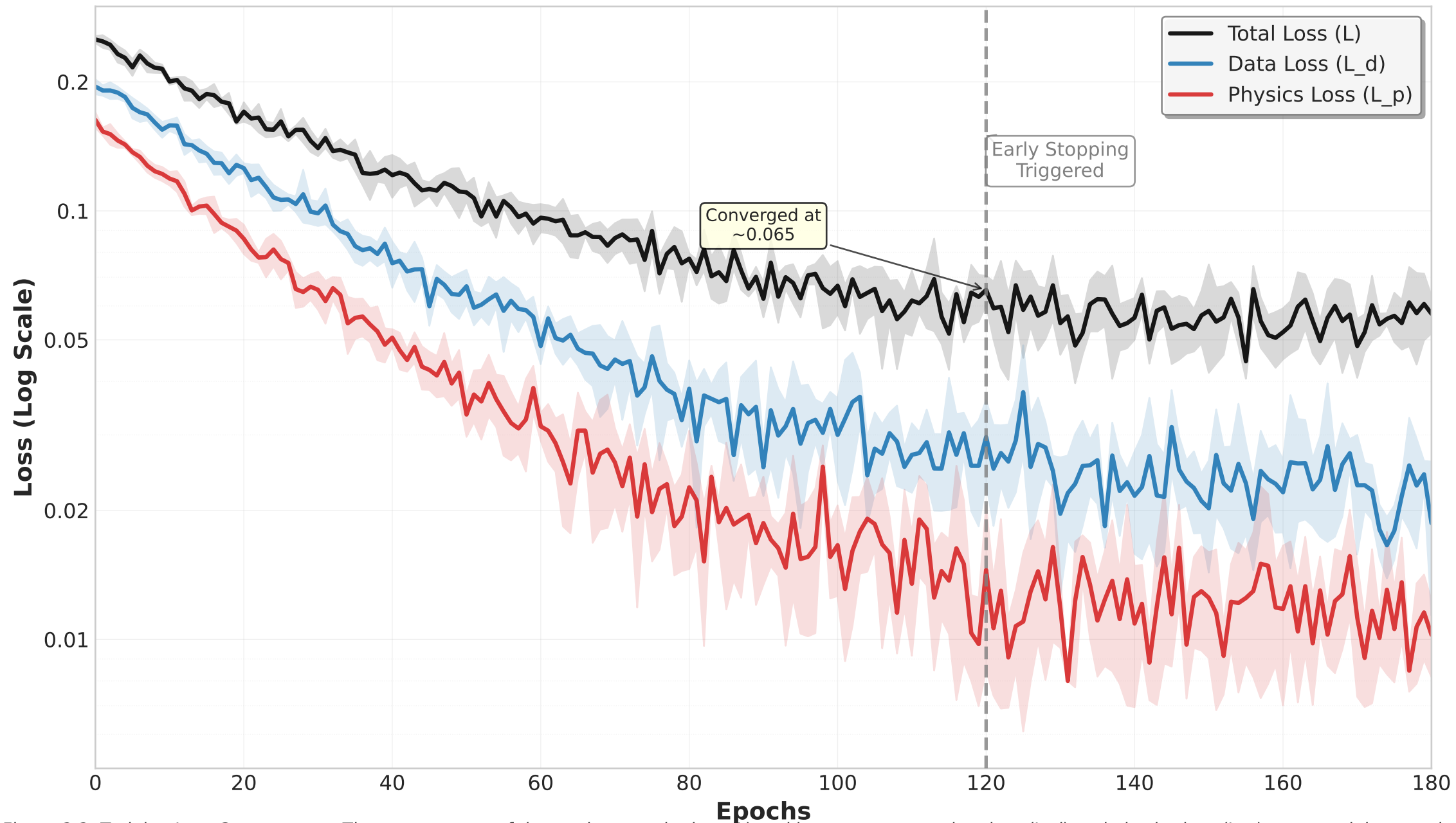


Figure 2.2: Training Loss Convergence. The convergence of the total composite loss (L) and its components—data loss (L_d) and physics loss (L_p)—over training epochs. The model shows rapid initial improvement followed by stable convergence. The vertical dashed line at epoch 120 indicates the point of early stopping, preventing overfitting. The shaded regions represent ± 1 standard deviation from 5 independent training runs, demonstrating the robustness of the training protocol and the reproducibility of the loss minimization.